



Rationalize the Consumption of Antibiotics

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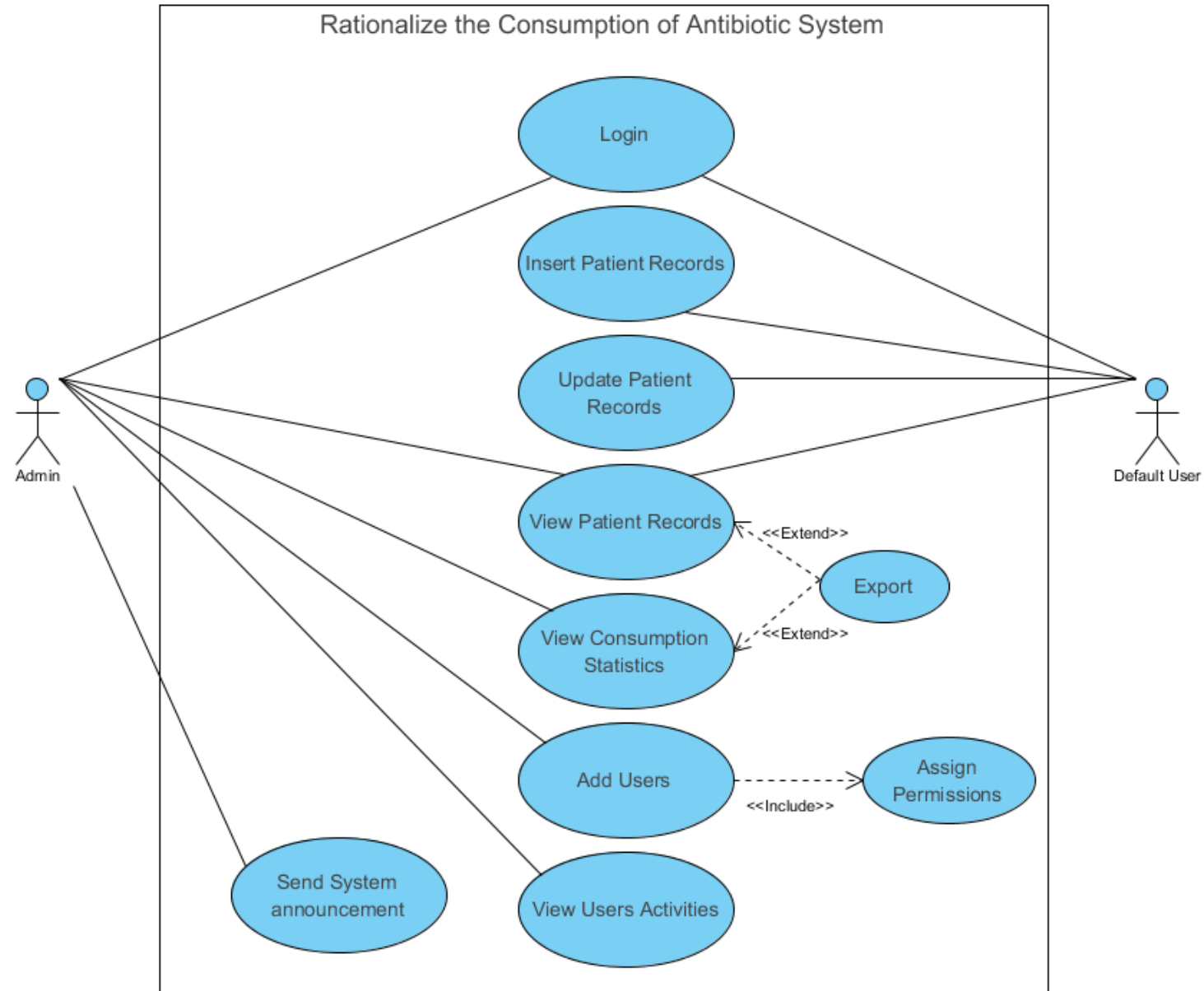
Problem Statement:

- There is a large number of antibiotics are being wrongly used daily.
- Each antibiotic can be used to treat more than one type of bacteria.
- The antibiotics required to treat a patient vary These doses vary according to the patient's age, gender, health condition and other factors.
- It is necessary to have a system followed in the consumption of antibiotics that aims to rationalize the consumption of antibiotics in their places of distribution, whether hospitals, clinics or pharmacies.

Project Objectives:

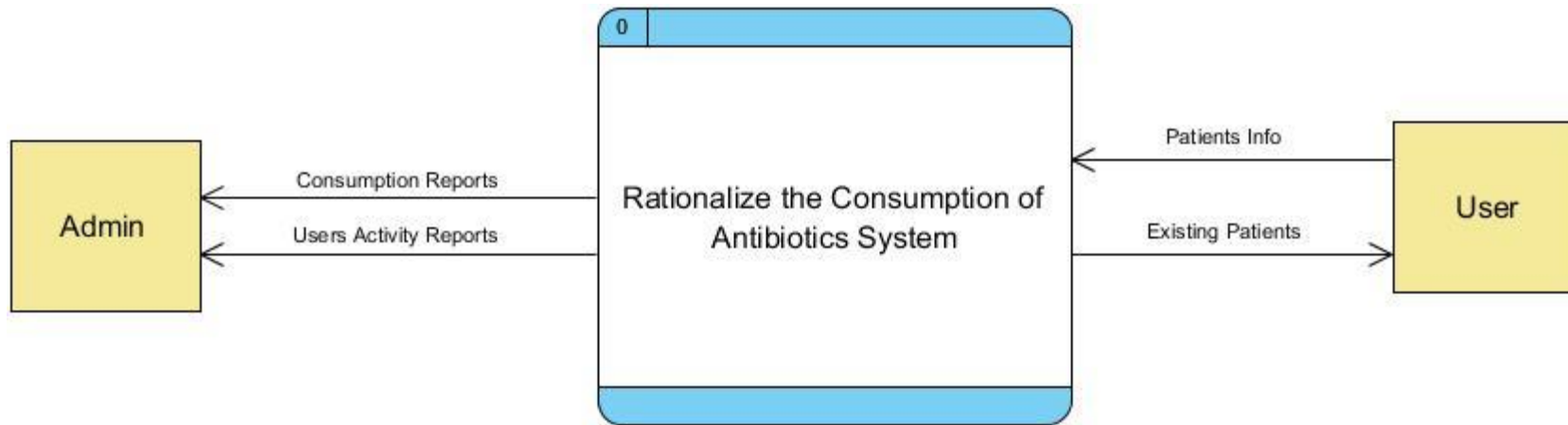
- The project aims to build a system that helps rationalize the consumption of antibiotics. The best way to rationalize consumption is to monitor the number of patients, the types of diseases they have, and the doses required for the treatment process.
- Objectives:
 - Insert/update Patient Records
 - Detecting if an antibiotic is being consumed inappropriately
 - Setting appropriate dosage instructions for each patient, according to the case, during the patient registration process
 - Monitoring and extracting patients' statistics at the hospital
 - Monitoring and extracting statistics of antibiotic consumption in the hospital

Usecase

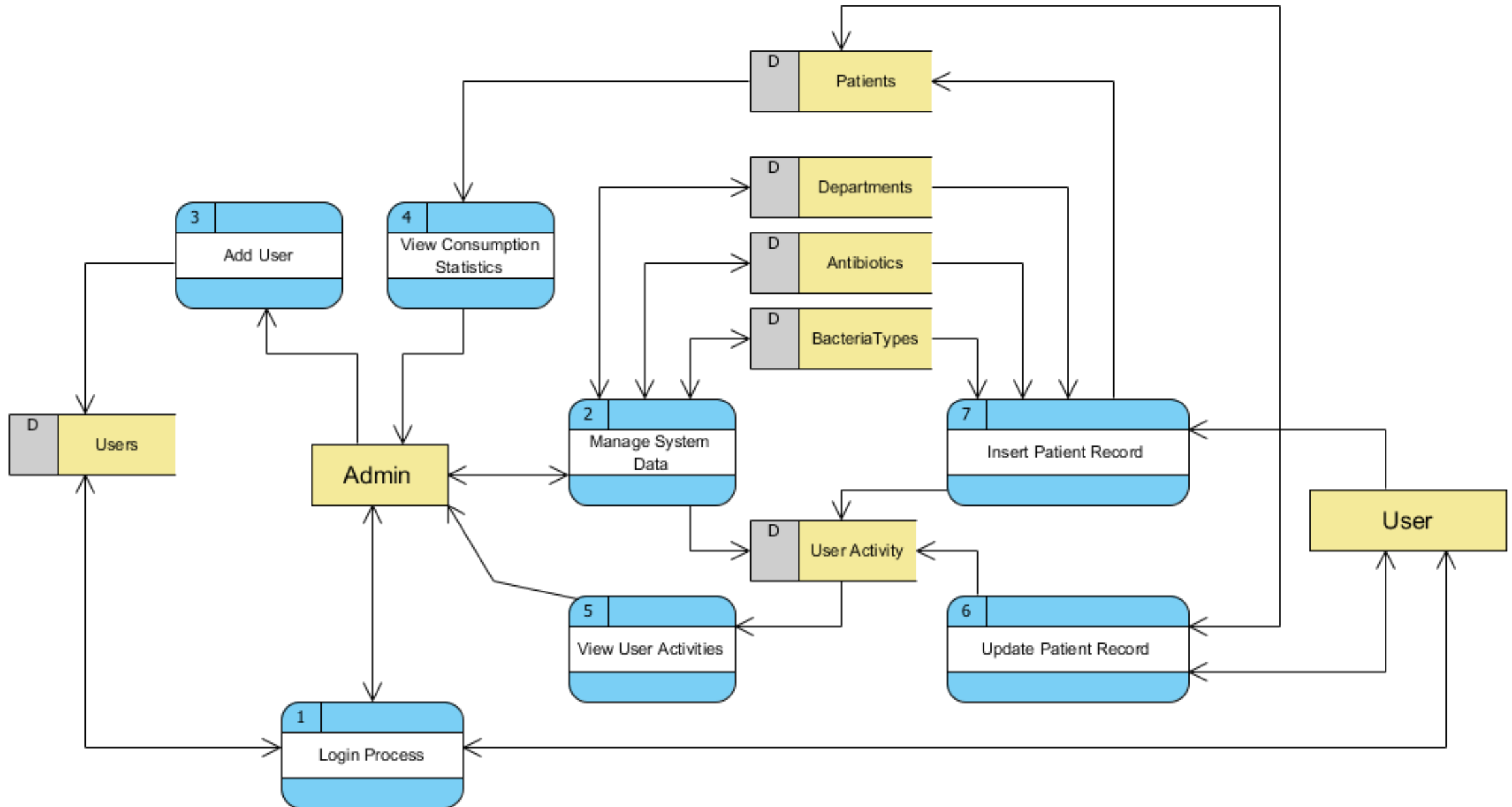


DFD Context

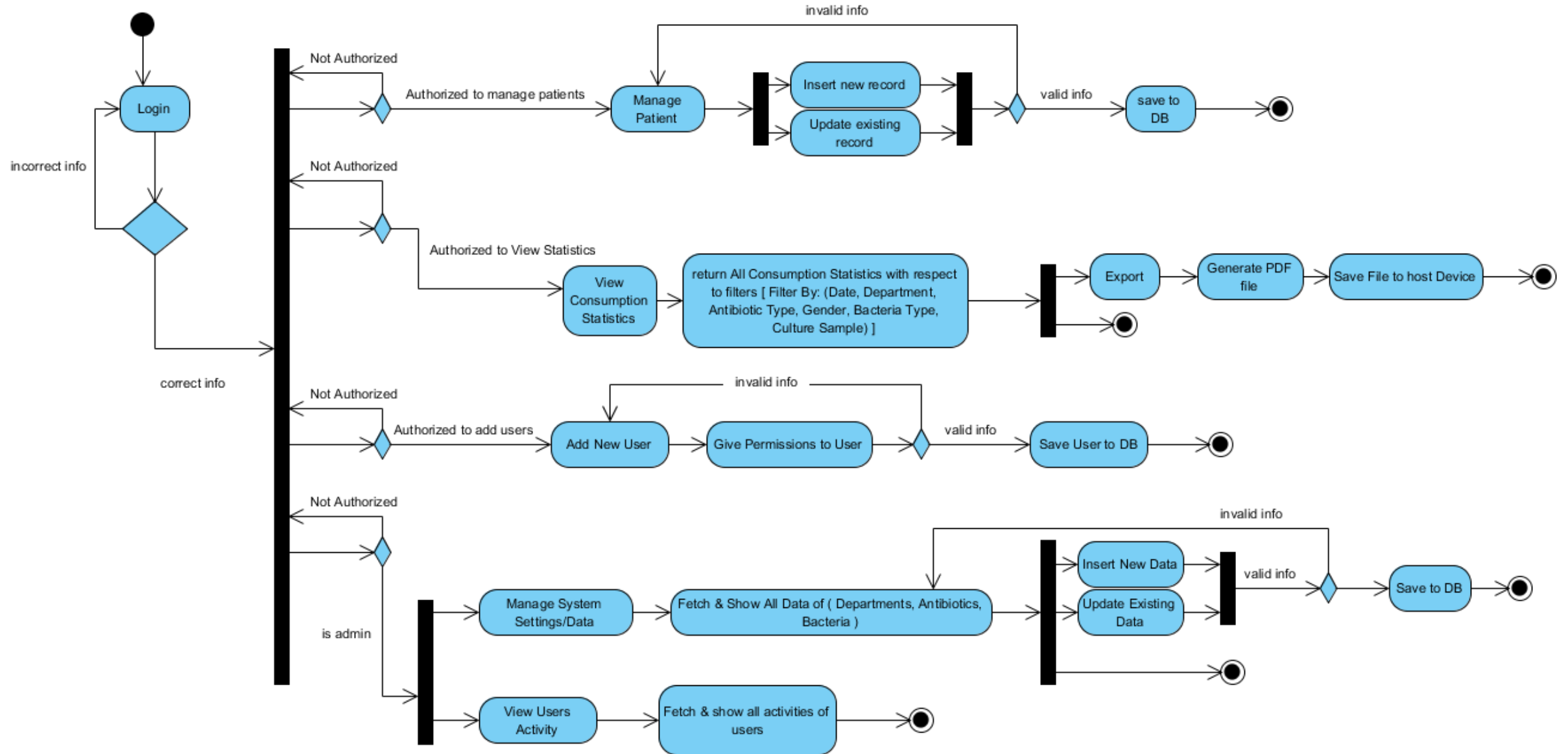
The base level of DFD on RCA Project shows the main input/output data on general bases, The user is responsible for inserting patients' data into the system, Whereas the admin can monitor these data and see reports



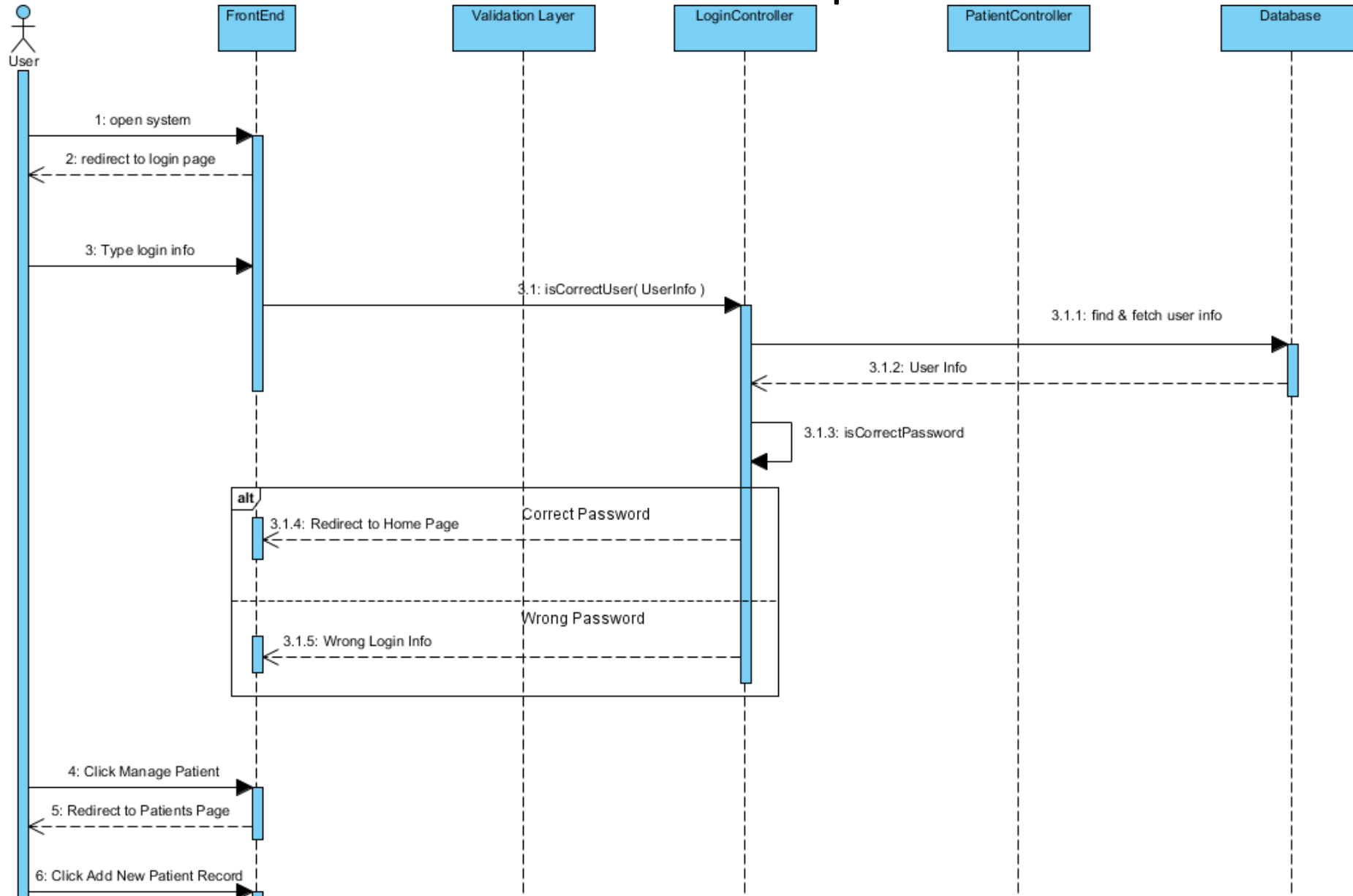
DFD Level 1



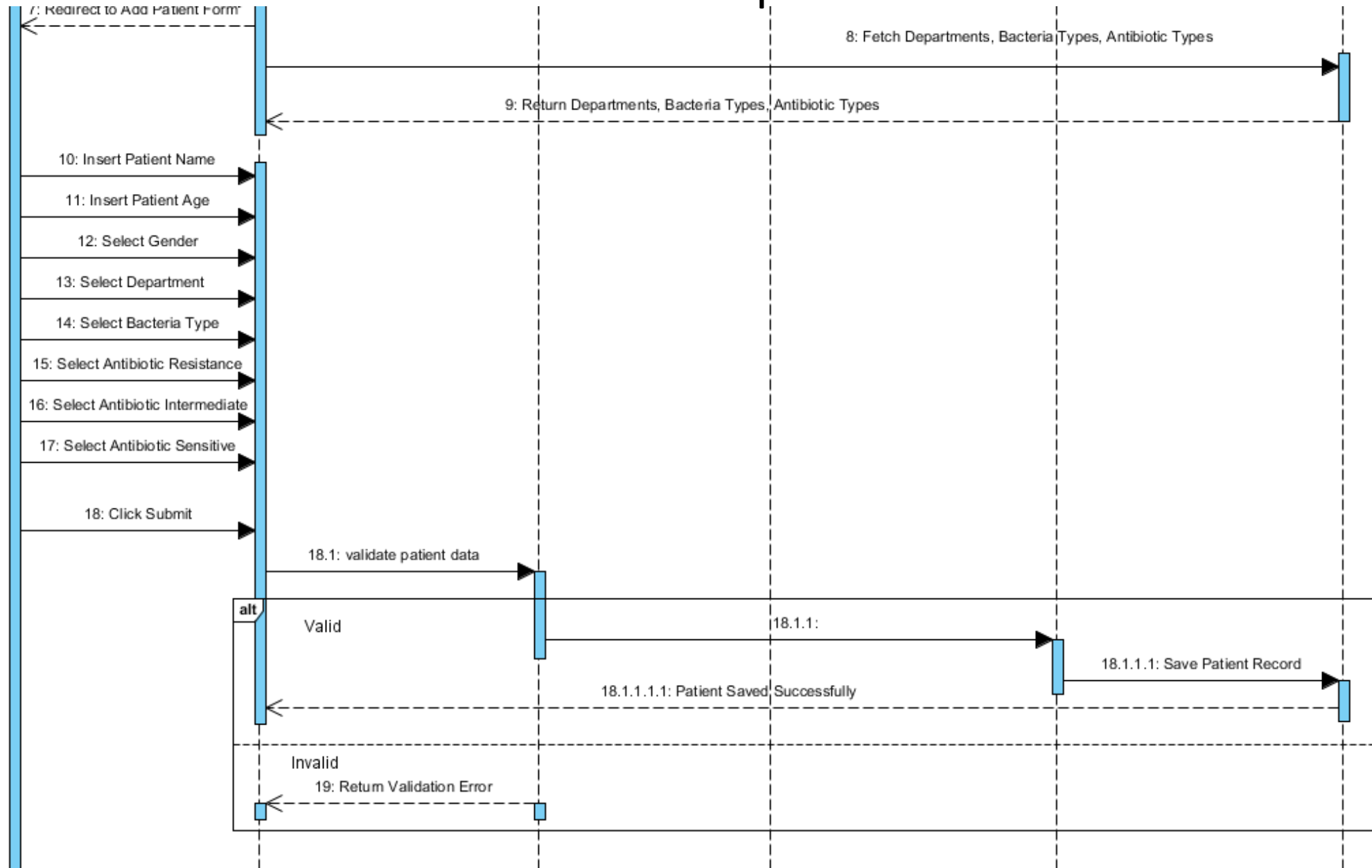
Activity



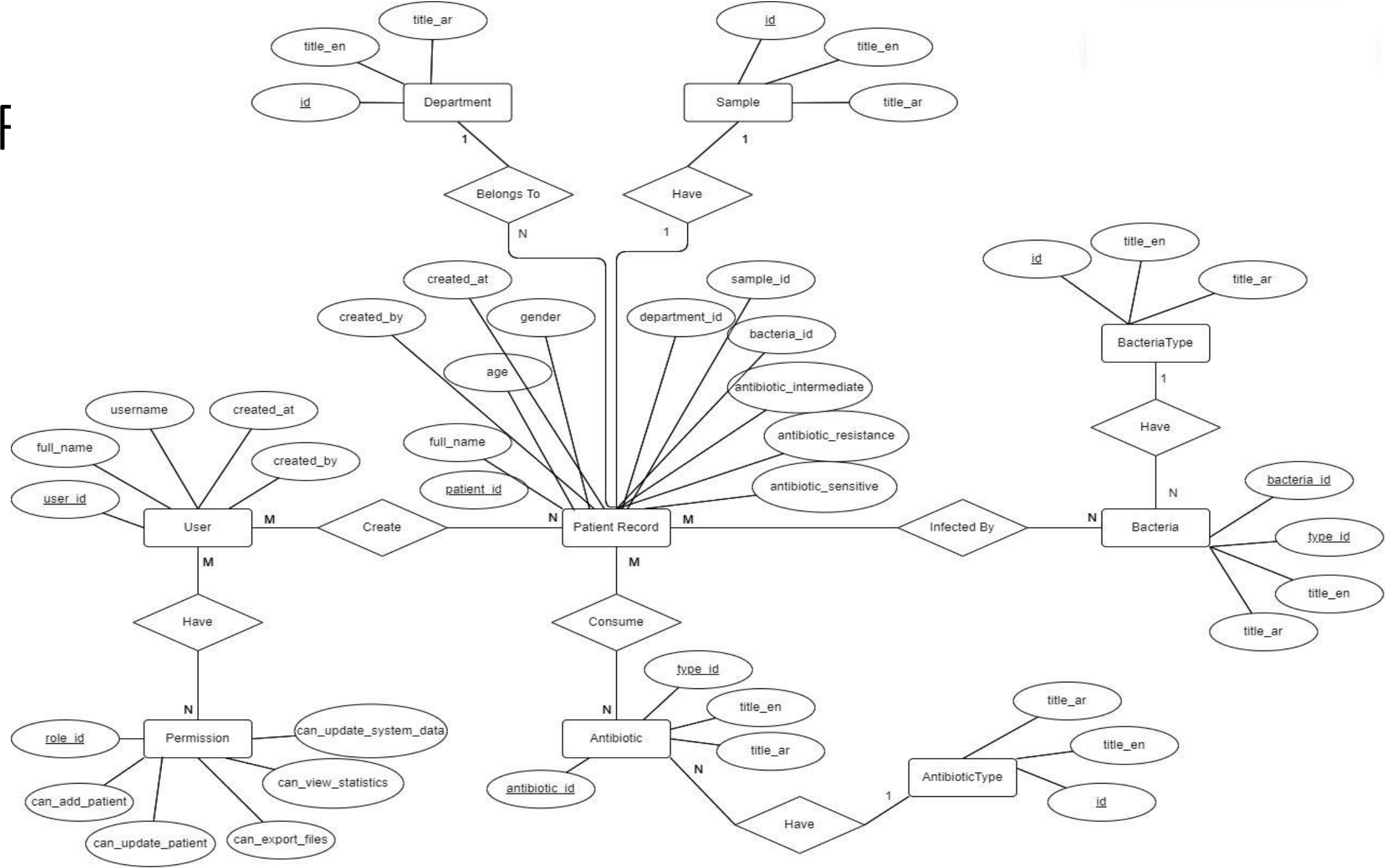
Insert Patient Sequence



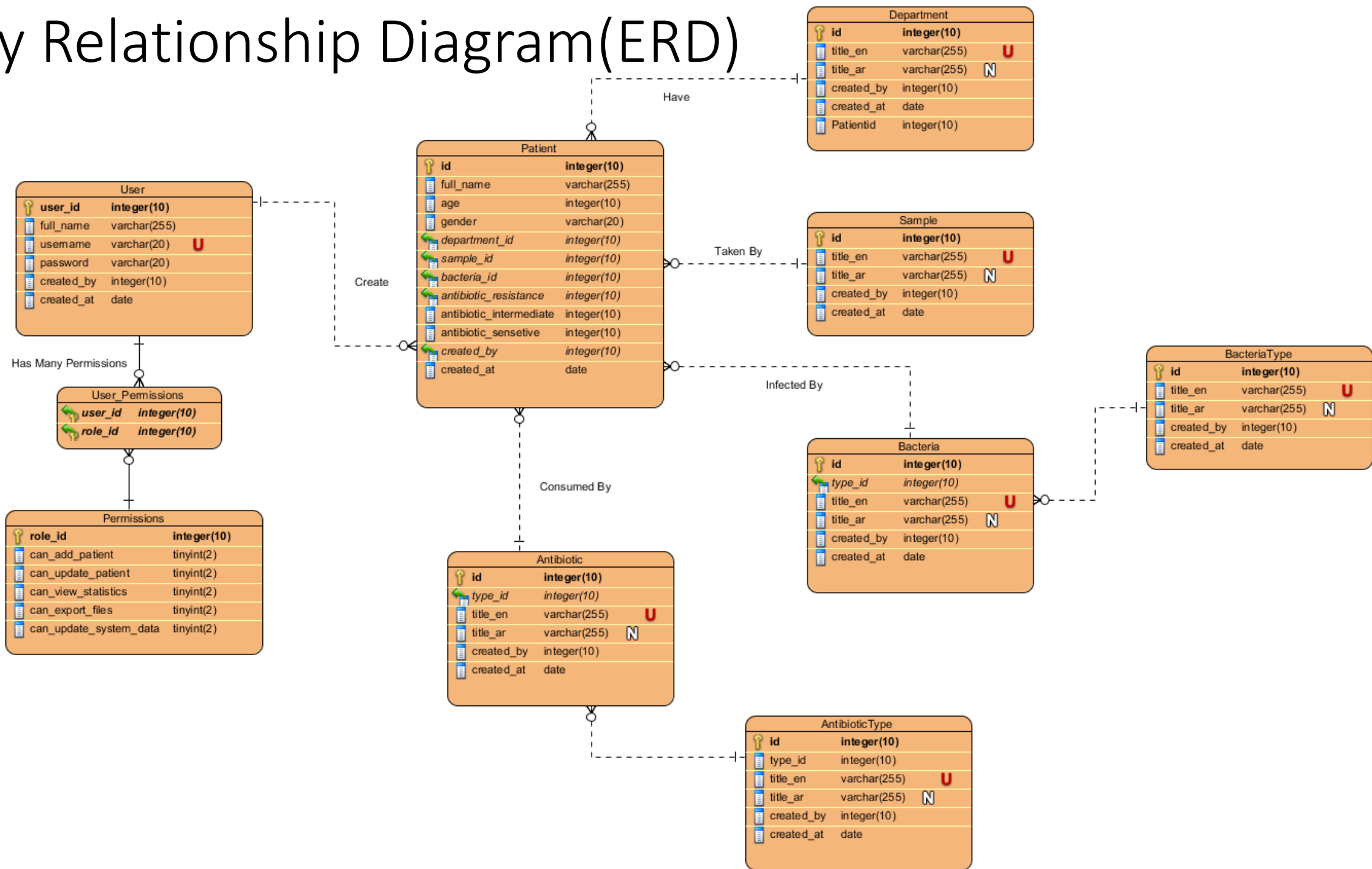
Insert Patient Sequence - Cont



EF



Entity Relationship Diagram(ERD)



APP Design

Login

Username admin

Password

☐ Remember Me

Login

RCA System

Not secure | rca.local/patients

Admin

Dashboard

Statistics

Manage Patients

Manage Users

Manage System Data

Settings

Patients Managements

New Patient Record

Search

search

id	Full Name	Bacteria	Antibiotic Resistance	Action
1	Mohamed Saleh Ibrahim	Staphylococcus aureus	amoxicillin	Edit
3	Sameh Ali Reyad	Klebsiella oxytoca	minocycline	Edit
17	Ismail Maohammed	Staphylococcus aureus	amoxicillin	Edit
18	Hend Salem Desoky	Staphylococcus aureus	amoxicillin	Edit
19	Maryam Abo EL Eila	Staphylococcus aureus	amoxicillin	Edit
20	Eman Khaled	Other coagulase-negative staphylococci (CNS)	dicloxacillin	Edit

RCA System

Not secure | rca.local/add-patient

Admin

Dashboard

Statistics

Manage Patients

Manage Users

Manage System Data

Settings

Add Patient Record

Name

Ahmed Mohamed Ibrahim

Age

25

Gender

Male

Department

General Surgery

Bacteria

Staphylococcus aureus

Sample

Blood culture

Antibiotic Resistance

amoxicillin

Intermediate Antibiotic

ampicillin

Antibiotic Sensetive

dicloxacillin

Save

RCA System

Not secure | rca.local/add-patient

Admin

Dashboard

Statistics

Manage Patients

Manage Users

Manage System Data

Settings

Add Patient Record

Name

Ahmed Mohamed Ibrahim

Age

25

Gender

Male

Department

General Surgery

Bacteria

Staphylococcus aureus

Lactobacillus spp.

Listeria monocytogenes

Gram-positive bacilli, not specified

Other gram-positive bacilli

Gram-positive bacilli

Citrobacter freundii

Citrobacter koseri

Citrobacter spp., other

Citrobacter spp., not specified

Enterobacter cloacae

Enterobacter aerogenes

Enterobacter agglomerans

Enterobacter sakazakii

Enterobacter spp., other

Enterobacter spp., not specified

Escherichia coli

Klebsiella pneumoniae

Klebsiella oxytoca

Klebsiella spp., other

Klebsiella spp., not specified

Antibiotic Resistance

amoxicillin

Intermediate Antibiotic

ampicillin

Antibiotic Sensitive

dicloxacillin

Save

RCA System

Not secure | rca.local/add-patient

Admin

Dashboard

Statistics

Manage Patients

Manage Users

Manage System Data

Settings

Add Patient Record

Name

Ahmed Mohamed Ibrahim

Age

25

Gender

Male

Department

General Surgery

Bacteria

Staphylococcus aureus

Sample

Blood culture

Antibiotic Resistance

amoxicillin

Intermediate Antibiotic

ampicillin

Penicillins

amoxicillin

amoxicillin and clavulanate

ampicillin

dicloxacillin

oxacillin

penicillin V potassium

Tetracyclines

demectocycline

doxycycline

eravacycline

minocycline

Cephalosporins

Fluoroquinolones

Lincomycins

Macrolides

Sulfonamides

Glycopeptide Antibiotics

Aminoglycosides

Save

Tools

Front-end :

- html5 , css3, JavaScript
- jQuery
- bootstrap
- flutter

Back-end :

- Laravel - Php
- SQL
- REST API

Version Control

- git

System Conclusion

- Admin can create many users with different permissions
- Users can Fill & Update patient's forms
- Admin or Authorized users can monitor consumption statistics
- Admin or Authorized users can send system announcements
- Admin can track users activities
- Admin can insert/update system data (Departments, Bacteria Types, Antibiotic Types)

Future Work

- Work on Statistics page, where users can monitor consumption statistics filtered by (Date, Departments, Antibiotic Type, Bacteria Type)
- Work on Mobile Application version of the system
- Work on system deployment

THANKS